

Q5. Analysing Ethical Issues in AI/ML Systems

1. Bias and Fairness

- **Bias in training data:** AI models learn patterns from historical data, which may contain social prejudices. For example, facial recognition systems have shown higher error rates for darker-skinned individuals.
- **Fairness in outcomes:** Algorithms used in hiring or lending must ensure equitable treatment across gender, race, and socio-economic backgrounds.
- **Mitigation strategies:** Diverse datasets, fairness metrics, and bias audits are essential to reduce discriminatory outcomes.

2. Privacy Concerns

- **Data collection:** AI systems often rely on massive amounts of personal data, raising risks of surveillance and misuse.
- **Consent and control:** Individuals may not fully understand how their data is used, undermining informed consent.
- **Regulatory frameworks:** Laws such as GDPR emphasize data protection, but enforcement remains inconsistent across regions.

3. Transparency and Explainability

- **Black-box problem:** Many ML models, especially deep learning, are opaque and difficult to interpret.
- **Need for explainability:** Stakeholders must understand how decisions are made, particularly in sensitive domains like healthcare or criminal justice.
- **Tools for transparency:** Explainable AI (XAI) techniques, model documentation, and algorithmic audits can improve accountability.

4. Accountability and Governance

- **Responsibility for harm:** When AI systems cause errors—such as wrongful arrests or medical misdiagnoses—determining accountability is complex.
- **Shared responsibility:** Developers, deployers, and regulators must collectively ensure ethical compliance.
- **Governance models:** Independent oversight bodies, ethical review boards, and international cooperation are needed to regulate AI deployment.

5. Case Studies and Real-World Examples

- **COMPAS algorithm:** Used in U.S. courts for predicting recidivism, criticized for racial bias.
- **Cambridge Analytica scandal:** Demonstrated how AI-driven profiling can manipulate political outcomes.
- **Healthcare AI:** While improving diagnostics, concerns remain about patient privacy and liability in case of errors.